Gravitational Lensing in Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)

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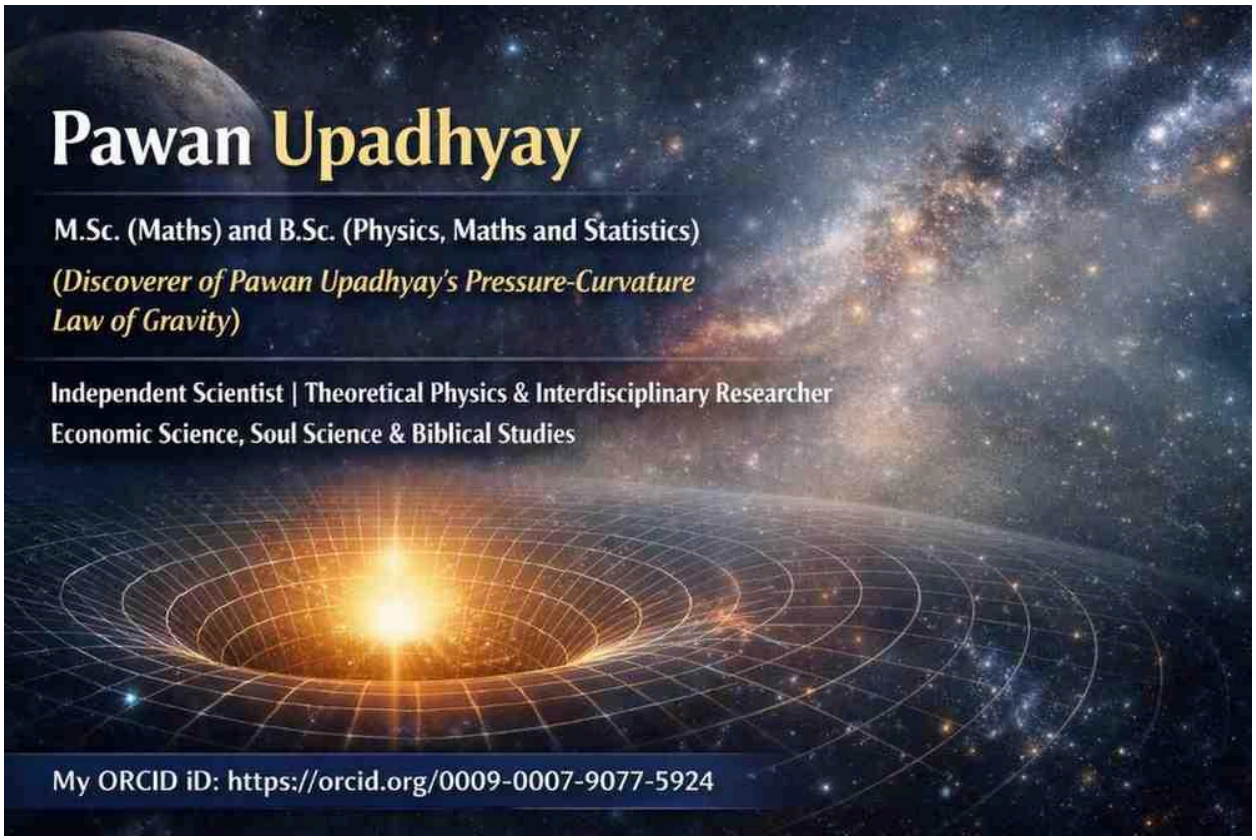

Abstract

Gravitational lensing is one of the most important observational confirmations of relativistic gravity. In General Relativity, light bending arises due to spacetime curvature generated by mass–energy. In this paper, we reinterpret gravitational lensing within the framework of Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity), where curvature is produced by gradients of gravitational pressure generated by mass–energy. We develop a

pressure-based formulation of light deflection, derive a PPC lens equation, and explore implications in weak and strong gravitational fields. The theory provides a physical mechanism for curvature and suggests possible deviations in extreme regimes.

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1. Introduction

Gravitational lensing refers to the bending of light in the presence of massive objects such as stars, galaxies, and black holes. Traditionally, this phenomenon is explained by General Relativity, where spacetime curvature guides the motion of light along null geodesics.

The PPC Law of Gravity introduces a new interpretation: curvature is not merely geometric but arises from gravitational pressure generated by mass–energy. This work applies the PPC framework to gravitational lensing and investigates how light propagation is governed by pressure-induced curvature.

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2. Conceptual Foundation of PPC Lensing

The PPC framework is based on the following sequence:

Mass Density (ρ) → Energy Density ($E_d = \rho c^2$) → Gravitational Pressure ($P_g = \omega E_d$) → Pressure Gradient (∇P_g) → Spacetime Curvature → Light Propagation

In this interpretation, gravitational lensing is not directly caused by mass, but by the pressure gradients produced by mass–energy distributions.

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3. Light Propagation in Curved Spacetime

Light follows null geodesics, meaning its trajectory is determined by spacetime geometry. In PPC gravity, this geometry is shaped by gravitational pressure gradients.

Thus, light behaves as a curvature-guided electromagnetic wave, revealing the structure of the pressure field in spacetime.

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4. Pressure–Curvature Relation

A central assumption of PPC gravity is that spacetime curvature can be decomposed into magnitude and geometric structure:

4.1 Curvature Magnitude

The strength of spacetime curvature is determined by the total effective energy contribution:

$$|C| \propto (E_d + P_g)$$

This shows that both energy density and gravitational pressure directly contribute to the magnitude of curvature.

4.2 Curvature Geometry

The geometric structure and directional properties of curvature are governed by pressure gradients:

$$C_{\mu\nu} \propto \nabla_\mu \nabla_\nu P_g$$

This indicates that spatial variations of gravitational pressure determine how spacetime bends and how light and matter propagate.

4.3 Physical Interpretation

Thus, PPC gravity provides a natural decomposition:

- $(E_d + P_g)$ determines the **magnitude** of curvature
- ∇P_g determines the **geometry** of curvature

This decomposition offers a clearer physical interpretation of spacetime curvature compared to standard formulations.

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5. Surface Density and PPC Deflection Angle

5.1 Surface Density Definition

Surface density is a key quantity in gravitational lensing, representing the projected mass distribution along the line of sight. It is defined as:

$$\Sigma(b) = \int \rho \, dl$$

where:

- $\Sigma(b)$ is the surface mass density at impact parameter b
- ρ is the mass density
- dl is the line-of-sight element

In the PPC framework, since gravitational pressure is related to energy density by $P_g = \omega E_d = \omega \rho c^2$, surface density can also be related to the projected pressure field.

5.2 PPC Deflection Angle

We define the deflection angle of light in PPC gravity as:

$$\alpha_{PPC} \propto \int \nabla_\perp P_g \, dl$$

where:

- $\nabla_\perp P_g$ is the pressure gradient perpendicular to the light path
- dl is the infinitesimal path element

This equation replaces the mass-based deflection in General Relativity with a pressure-gradient-based formulation.

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This equation replaces the mass-based deflection in General Relativity with a pressure-gradient-based formulation.

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6. Weak-Field Approximation

In weak gravitational fields, where pressure gradients are small, PPC predictions should reduce to those of General Relativity. In this limit, gravitational pressure is proportional to energy density, and curvature behaves similarly to standard relativistic predictions.

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7. Strong-Field Regime

In extreme environments such as neutron stars and black holes, pressure becomes significant. In these cases:

∇P_g becomes large

This suggests that gravitational lensing may depend strongly on internal pressure distributions, potentially leading to:

- Enhanced light deflection
- Modified Einstein ring structures
- Asymmetric lensing patterns

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8. Gravitational Pressure Waves and Lensing

In PPC gravity, gravitational waves are interpreted as dynamic pressure disturbances. These disturbances can modify spacetime curvature dynamically, potentially affecting light propagation and leading to time-dependent lensing effects.

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9. Comparison with General Relativity

Feature	General Relativity	PPC Law
Source of curvature	Mass–energy tensor	Pressure + energy
Cause of lensing	Spacetime curvature	Pressure-gradient-induced curvature
Deflection dependence	Mass distribution	Pressure gradient distribution

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10. Observational Implications

The PPC framework suggests that gravitational lensing may reveal information about pressure distributions in astrophysical objects. Potential tests include:

- Lensing by neutron stars
- Strong lensing near black holes
- Cosmological lensing in high-pressure early universe conditions

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11. Discussion

The PPC interpretation provides a physical mechanism underlying spacetime curvature. While mathematically consistent with relativistic frameworks, it emphasizes pressure as the key driver of curvature.

Further work is required to derive exact solutions and compare predictions quantitatively with observations.

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12. Conclusion

Gravitational lensing in PPC gravity arises from pressure-gradient-induced curvature of spacetime. Light follows paths determined by this curvature, acting as a probe of the gravitational pressure field.

The theory offers a new perspective on gravity and suggests possible deviations in extreme conditions, providing opportunities for future observational tests.

Keywords

Gravitational Lensing, Pressure Curvature Law, PPC Gravity, Spacetime Curvature, Light Deflection, Theoretical Physics

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Gravitational Lensing in Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)

Author: Pawan Upadhyay

Field: Theoretical Physics – Gravitation

Abstract

Gravitational lensing is a fundamental observational phenomenon confirming the existence of spacetime curvature. In standard gravitational theory, light deflection arises due to curvature generated by mass–energy. In the Pressure Curvature Law of Gravity (PPC), curvature is interpreted as a consequence of gravitational pressure generated by energy density. This paper presents a comprehensive formulation of gravitational lensing within the PPC framework, including curvature decomposition, geodesic motion, pressure-driven deflection, and a modified Einstein radius. The theory provides a physically intuitive mechanism for gravity and predicts deviations from standard relativistic lensing in strong-field regimes.

1. Introduction

Gravitational lensing refers to the bending of light in the presence of massive astrophysical objects such as stars, galaxies, and black holes. In conventional theory, this phenomenon is explained through spacetime curvature, where light follows null geodesics determined by geometry.

The PPC Law of Gravity introduces an alternative interpretation: spacetime curvature arises from gravitational pressure generated by mass–energy. This framework provides a physical mechanism underlying curvature and allows gravitational lensing to be described in terms of pressure gradients.

2. PPC Framework

The PPC formulation follows the sequence:

$$\rho \to E_d = \rho c^2 \to P_g = \omega E_d \to \nabla P_g \to \text{Curvature} \to \text{Light propagation}$$

Thus, gravitational phenomena arise from the distribution and variation of gravitational pressure.

3. Light Propagation in PPC Gravity

Light propagates along null geodesics determined by spacetime curvature. In PPC gravity, curvature is generated by pressure gradients; therefore, light follows trajectories shaped by the gravitational pressure field.

Light behaves as a **curvature-guided electromagnetic wave**, revealing the structure of the pressure field.

4. Pressure–Curvature Decomposition

4.1 Curvature Magnitude

$$|C| \propto (E_d + P_g)$$

Energy density and gravitational pressure determine the strength of curvature.

4.2 Curvature Geometry

$$C_{\mu\nu} \propto \nabla_\mu \nabla_\nu P_g$$

Pressure gradients determine the structure and direction of curvature.

4.3 Interpretation

- $E_d + P_g$: magnitude of curvature
- ∇P_g : geometry of curvature

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5. Geodesic Motion in PPC Gravity

The motion of particles is governed by:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}(E_d, P_g)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0$$

where the affine connection depends on energy density and gravitational pressure. This formulation preserves relativistic covariance while attributing curvature to physical pressure fields.

6. Pressure-Induced Force for Extended Bodies

For extended bodies, gravitational pressure acts on the surface:

$$\vec{F} = \oint P_g \hat{n} \, dA$$

Using the divergence theorem:

$$\vec{F} = -\int \nabla P_g \, dV$$

Thus, gravitational motion arises from pressure imbalance across the body.

7. PPC Deflection of Light

The deflection of light is governed by pressure gradients:

$$\alpha_{PPC} = k \int \nabla_{\perp} P_g \, dl$$

This replaces mass-based deflection with a pressure-gradient formulation.

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8. Mathematical Formulation

8.1 Weak-Field Approximation

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad h_{\mu\nu} \propto P_g$$

Define effective potential:

$$\Phi_{PPC} \propto (E_d + P_g)$$

8.2 Light Deflection Equation

$$\frac{d^2 x_\perp}{dz^2} \approx -\nabla_\perp \Phi_{PPC}$$

Dominant contribution:

$$\frac{d^2 x_\perp}{dz^2} \propto -\nabla_\perp P_g$$

9. PPC Einstein Radius

Starting from:

$$\alpha_{PPC}(b) = k \int \nabla_{\perp} P_g \, dl$$

Using:

$$P_g = \omega \rho c^2$$

Surface density:

$$\Sigma(b) = \int \rho \, dl$$

For a point mass:

$$\alpha_{PPC}(b) \propto \frac{k\omega c^2 M}{b^3}$$

Lens equation:

$$\theta_E D_s = \alpha_{PPC}(\theta_E) D_{ls}$$

Final PPC Einstein radius:

$$\boxed{\theta_E^{PPC} = \left(\frac{k\omega c^2 M D_{ls}}{D_l^3 D_s} \right)^{1/4}}$$

10. Comparison with General Relativity

Feature	Standard Gravity
Driver	Mass–energy
Deflection	$\propto M/b$
Einstein radius	$\propto M^{1/2}$

Standard Gravity	PPC Gravity
Mass–energy	Pressure gradients
$\propto M/b$	$\propto 1/b^3$
$\propto M^{1/2}$	$\propto M^{1/4}$

11. Strong-Field Predictions

In extreme regimes (e.g., neutron stars, black holes):

- Enhanced dependence on pressure
- Modified lensing structures
- Deviations from standard relativistic predictions

12. Observational Implications

- Lensing by compact stars
- Strong lensing near black holes
- Early universe high-pressure lensing

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13. Conclusion

Gravitational lensing in PPC gravity arises from pressure-gradient-induced curvature. The theory provides a physically intuitive mechanism for gravity and predicts measurable deviations from standard gravitational lensing in extreme conditions.

Keywords

Gravitational Lensing, PPC Gravity, Pressure Curvature, Einstein Radius, Light Deflection, Theoretical Physics

🧠 Two Forces in PPC

🔹 1. Field (Body) Force

$$\vec{F}_{field} = -\nabla P_g$$

Meaning:

- Acts **throughout the volume**
- Comes from **spatial variation of pressure**
- Drives motion of particles and fluids

🔹 2. Surface Force

$$F_p = P_g A$$

More generally:

$$\vec{F}_p = \oint P_g \, \hat{n} \, dA$$

Meaning:

- Acts **on the boundary of an extended body**
- Depends on **local pressure at the surface**
- Describes how pressure “pushes” on surfaces

⚖️ Key Difference (Very Important)

Field Force	Surface Force
Volume force	Boundary force
Gradient ∇P_g	Value P_g
Point / volume	Surface
Causes acceleration	Describes pressure interaction

They are mathematically linked by:

$$\boxed{\vec{F}_{surface} = \oint P_g \hat{n} dA = -\int \nabla P_g \, dV}$$

👉 This means:

- **Surface force = integrated effect of field force**
- If pressure varies → both give the same net result

🌌 PPC Interpretation

✔ Field force

> “Gravity is caused by pressure gradients in spacetime”

✔ Surface force

> “Gravity appears as pressure acting on extended bodies”

👉 So:

- Field force → **fundamental law**
- Surface force → **macroscopic manifestation**

Field and Surface Forces in PPC Gravity

In PPC gravity, gravitational interaction can be described in two equivalent forms. The local field force is given by the pressure gradient:

$$\vec{F}_{field} = -\nabla P_g$$

For extended bodies, gravitational pressure acts on the surface:

$$\vec{F}_{surface} = \oint P_g \hat{n}\, dA$$

These two descriptions are related through:

$$\vec{F}_{surface} = -\int \nabla P_g\, dV$$

Thus, surface forces represent the integrated effect of pressure gradients across the volume of the body.

$$\boxed{\oint P_g \hat{n}\, dA = -\int \nabla P_g \, dV}$$

👉 This is the **divergence theorem**

If pressure is **uniform**:

$$P_g = \text{constant}$$

👉 Then:

- Forces cancel
- Net force = 0

If pressure is **non-uniform**:

$$\nabla P_g \neq 0$$

👉 Then:

- Net force appears
- Body accelerates